Problem 1

The ABB arm has the following DH table.

Link	a	d	α	θ
1	0	d_1	$\pi/2$	$\star$
2	a_2	0	0	$\star$
3	a_3	0	$-\pi/2$	$\star$
4	0	d_4	$\pi/2$	$\star$
5	0	0	$-\pi/2$	$\star$
6	0	d_6	0	$\star$

(a) Draw the zero-angle configuration.

(b) Solve the inverse kinematics (show all your work).

(c) How many possible solutions exist?

(d) Write a MATLAB code to demonstrate the efficacy of your script over 100 random test cases.

Part (a)

Evaluating the DH table with all $\theta_i = 0$ (see `Problem1Figure.m`, Appendix A):

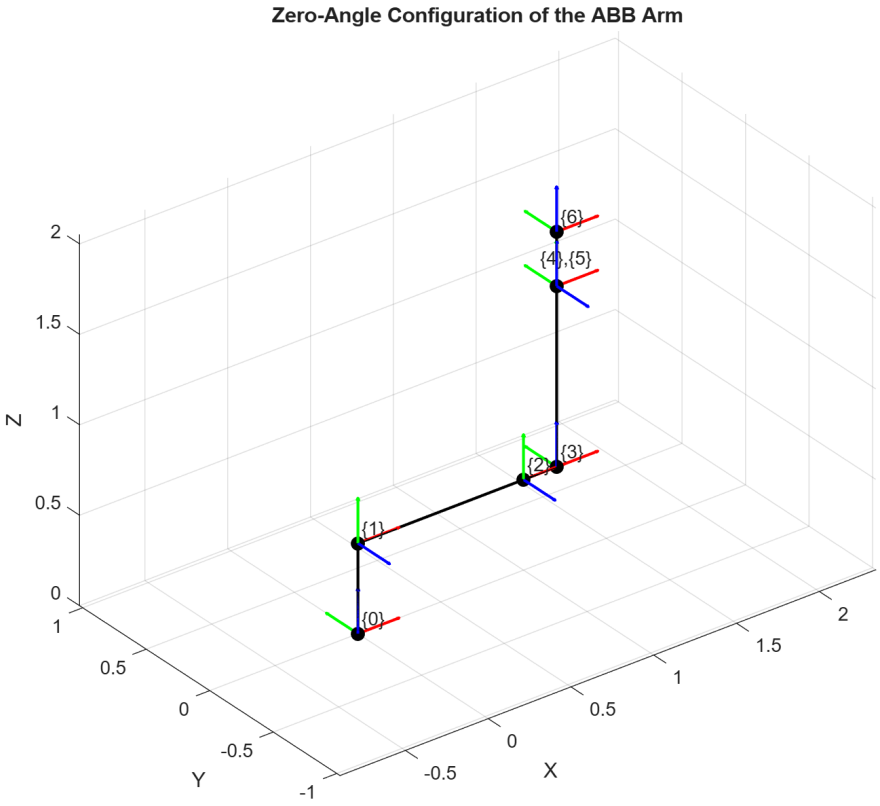


Figure 1: Zero-angle configuration of the ABB arm. Axes: x red, y green, z blue. Dimensions are illustrative and differ from Part (d) ($d_1 = 0.5$, $a_2 = 1$, $a_3 = 0.2$, $d_4 = 1$, $d_6 = 0.3$). Frames 4 and 5 share an origin, so they carry one combined label.

Part (b)

The last three axes intersect at the wrist center (origin of frames {4} and {5}), so the position and orientation problems decouple. Let $\boldsymbol{P}_{06} = [P_{6x},\, P_{6y},\, P_{6z}]^{\mathsf{T}}$ and let $\boldsymbol{R} = {}^{0}\boldsymbol{R}_{6}$ be the desired orientation. The wrist center is

$$\boldsymbol{P}_{05} = \begin{bmatrix} P_{5x} \\ P_{5y} \\ P_{5z} \end{bmatrix} = \boldsymbol{P}_{06} - d_6\, \boldsymbol{R}(:,3). \tag{1}$$

Joint 1. The projection of $\boldsymbol{P}_{05}$ onto the x_0–y_0 plane lies at angle θ_1 from x_0, so

$$\theta_1 = \operatorname{atan2}(P_{5y},\, P_{5x}) \quad \text{or} \quad \theta_1 + \pi. \tag{2}$$

Joints 2 and 3. With θ_1 known, the arm reduces to a planar two-link problem in the plane of $\boldsymbol{P}_{05}$. Define

$$r = \sqrt{P_{5x}^2 + P_{5y}^2}, \tag{3}$$

$$D = \sqrt{P_{5x}^2 + P_{5y}^2 + (P_{5z} - d_1)^2}, \tag{4}$$

$$L = \sqrt{a_3^2 + d_4^2}, \tag{5}$$

where D is the distance from O_1 to the wrist center and L is the distance from O_2 to the wrist center. The pose is reachable only if $|a_2 - L| \le D \le a_2 + L$. Let ψ be the elevation angle of D above the horizontal and $\beta \in [0, \pi]$ the interior angle at O_1 between a_2 and D. For θ_1 the horizontal reach is r; for $\theta_1 + \pi$ the arm points the other way, so r is replaced by $-r$:

$$\psi = \operatorname{atan2}(P_{5z} - d_1,\, \pm r) \quad (+r \text{ for } \theta_1,\ -r \text{ for } \theta_1 + \pi), \qquad \theta_2 = \psi \pm \beta. \tag{6}$$

Law of cosines on the triangle with sides a_2, D, L:

$$L^2 = a_2^2 + D^2 - 2a_2 D\cos\beta, \tag{7}$$

$$\cos\beta = \frac{a_2^2 + D^2 - L^2}{2a_2 D}, \qquad \sin\beta = +\sqrt{1 - \cos^2\beta}, \tag{8}$$

$$\beta = \operatorname{atan2}\left(\sqrt{1 - \left(\tfrac{a_2^2+D^2-L^2}{2a_2 D}\right)^2},\ \tfrac{a_2^2+D^2-L^2}{2a_2 D}\right). \tag{9}$$

The only sign choice is the $\pm$ in $\theta_2 = \psi \pm \beta$:

$$\theta_2 = \operatorname{atan2}(P_{5z} - d_1,\, \pm r) \pm \operatorname{atan2}\left(\sqrt{1 - \left(\tfrac{a_2^2+D^2-L^2}{2a_2 D}\right)^2},\ \tfrac{a_2^2+D^2-L^2}{2a_2 D}\right). \tag{10}$$

For θ_3, let δ be the fixed angle at O_2 between a_3 (the segment $O_2 \to O_3$) and L (the segment $O_2 \to$ wrist center), and γ the interior angle at O_2 between a_2 and L:

$$\delta = \operatorname{atan2}(d_4,\, a_3)\,. \tag{11}$$

Law of cosines on the same triangle, now for the angle opposite D:

$$D^2 = a_2^2 + L^2 - 2a_2 L\cos\gamma, \tag{12}$$

$$\cos\gamma = \frac{a_2^2 + L^2 - D^2}{2a_2 L}, \qquad \sin\gamma = +\sqrt{1 - \cos^2\gamma}, \tag{13}$$

$$\gamma = \operatorname{atan2}\left(\sqrt{1 - \cos^2\gamma},\ \cos\gamma\right). \tag{14}$$

The elbow sign of θ_3 is tied to the sign of β in θ_2:

$$\theta_3 = \mp(\pi - \gamma) - \delta, \tag{15}$$

with the upper sign paired with $\theta_2 = \psi + \beta$ (elbow up) and the lower sign with $\theta_2 = \psi - \beta$ (elbow down). Explicitly,

$$\theta_3 = \mp\left[\pi - \operatorname{atan2}\left(\sqrt{1 - \left(\tfrac{a_2^2+L^2-D^2}{2a_2L}\right)^2},\ \tfrac{a_2^2+L^2-D^2}{2a_2L}\right)\right] - \operatorname{atan2}(d_4,\, a_3)\,. \tag{16}$$

Joints 4, 5, and 6. From the DH table, ${}^0\boldsymbol{R}_3$ depends only on $\theta_1, \theta_2, \theta_3$:

$${}^0\boldsymbol{R}_3 = \operatorname{Rot}_z(\theta_1)\ \operatorname{Rot}_x(\tfrac{\pi}{2})\ \operatorname{Rot}_z(\theta_2+\theta_3)\ \operatorname{Rot}_x(-\tfrac{\pi}{2}) = \operatorname{Rot}_z(\theta_1)\ \operatorname{Rot}_y(-(\theta_2+\theta_3)). \tag{17}$$

Then

$${}^3\boldsymbol{R}_6 = \left({}^0\boldsymbol{R}_3\right)^{\mathsf{T}}\,{}^0\boldsymbol{R}_6. \tag{18}$$

From the DH table, the wrist rotation is ${}^3\boldsymbol{R}_6 = \operatorname{Rot}_z(\theta_4)\ \operatorname{Rot}_y(-\theta_5)\ \operatorname{Rot}_z(\theta_6)$. This is a Z–Y–Z rotation with θ_5 measured about $-y$, so the signs differ from the textbook form. Multiplying out, with r_{ij} the entries of ${}^3\boldsymbol{R}_6$,

$$r_{13} = -\cos\theta_4 \sin\theta_5, \qquad r_{23} = -\sin\theta_4 \sin\theta_5, \qquad r_{33} = \cos\theta_5, \tag{19}$$

$$r_{31} = \sin\theta_5 \cos\theta_6, \qquad r_{32} = -\sin\theta_5 \sin\theta_6. \tag{20}$$

Solving for the angles:

$$\theta_5 = \operatorname{atan2}\left(\pm\sqrt{r_{13}^2 + r_{23}^2},\ r_{33}\right), \tag{21}$$

$$\theta_4 = \operatorname{atan2}\left(\frac{-r_{23}}{\sin\theta_5},\ \frac{-r_{13}}{\sin\theta_5}\right), \tag{22}$$

$$\theta_6 = \operatorname{atan2}\left(\frac{-r_{32}}{\sin\theta_5},\ \frac{r_{31}}{\sin\theta_5}\right). \tag{23}$$

Singularities.

- *Shoulder* ($P_{5x} = P_{5y} = 0$): the wrist center lies on the z_0 axis and θ_1 is free.
- *Elbow* ($D = a_2 + L$ or $D = |a_2 - L|$): $\beta = 0$ (and $\gamma = 0$ or π), so the two elbow branches merge.
- *Wrist* ($\sin\theta_5 = 0$): axes 4 and 6 align and only $\theta_4 \pm \theta_6$ is determined. Choose $\theta_4 = 0$, and take θ_6 from the upper-left 2×2 block of ${}^3\boldsymbol{R}_6$: $\theta_6 = \operatorname{atan2}(r_{21}, r_{11})$ for $\theta_5 = 0$ and $\theta_6 = \operatorname{atan2}(r_{21}, -r_{11})$ for $\theta_5 = \pi$.

Part (c)

Up to 8 solutions for a reachable pose.

The count is $2 \times 2 \times 2$, one factor for each of the three sign choices:

- Shoulder: $\theta_1 = \operatorname{atan2}(P_{5y}, P_{5x})$ or $+\pi$ (front or back), giving 2 options.

- Elbow: the $\pm\beta$ sign in θ_2, with the matching $\mp(\pi-\gamma)$ in θ_3 (elbow up or down), giving 2 options for each shoulder choice.
- Wrist: the $\pm$ on $\sin\theta_5$ (wrist flip), giving 2 options for each arm configuration.

In degenerate cases the count changes. At the elbow singularity the two elbow branches coincide, and at the shoulder singularity θ_1 is free (infinitely many solutions). At the wrist singularity θ_4 and θ_6 are not separately determined, so each arm configuration has infinitely many wrist solutions. Outside the reachable range $|a_2 - L| \le D \le a_2 + L$ there are none.

Part (d)

See `Problem1.m` and `ikABB.m`, Appendix A. The test uses $d_1 = 0.290$, $a_2 = 0.270$, $a_3 = 0.070$, $d_4 = 0.302$, $d_6 = 0.072$ (m), seed 544, and a tolerance of 10^{-8}. Each of 100 random joint vectors has $\theta_1, \ldots, \theta_4, \theta_6 \in [-\pi, \pi]$ and $\theta_5 = \pm(0.05 + 0.9u)\pi$ with $u \sim U[0,1]$ and a random sign, which excludes a band of 0.05π around the wrist singularity and exercises both wrist branches. Forward kinematics gives the target pose $\boldsymbol{T}_{06}$; `ikABB` returns all 8 solutions, and each is passed through forward kinematics and compared to $\boldsymbol{T}_{06}$. Position and rotation errors are reported separately, and the distinctness row shows that the 8 solutions are pairwise different. One separate singular case ($\theta_5 = 0$) checks the wrist-singularity branch.

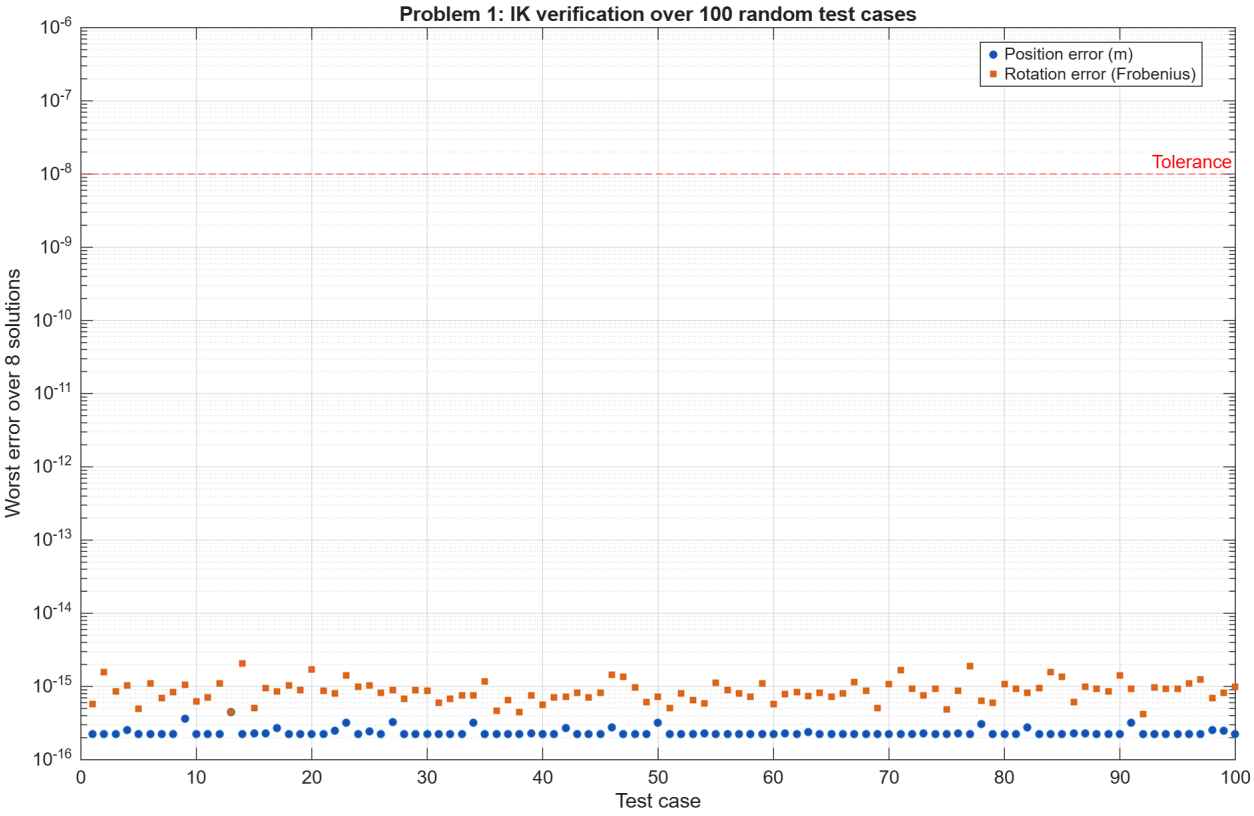


Figure 2: Worst position and rotation error over the 8 IK solutions for each of the 100 random test cases.

Appendix A: Code

Problem 1

Listing 1: Problem1.m

```
% Problem1: Inverse kinematics of the ABB arm, tested over random poses
%
% Arael Anaya
% 10920967
% MEGN544
% 10/03/2026

clear; close all; clc;

addpath(genpath('../../../Functions'));

rng(544);
numTests = 100;
tol = 1e-8;
distTol = 1e-6;

d1 = 0.290;  a2 = 0.270;  a3 = 0.070;  d4 = 0.302;  d6 = 0.072;
dims = [d1 a2 a3 d4 d6];

DH = [      0     d1     pi/2      0    ;
            a2    0        0       0    ;
            a3    0     -pi/2      0    ;
            0     d4     pi/2      0    ;
            0     0     -pi/2      0    ;
            0     d6       0       0  ];

numPoseOK = 0;
numRecovered = 0;
numDistinct = 0;
worstPosErr = 0;
worstRotErr = 0;
minPairDist = inf;
testPosErr = zeros(numTests, 1);
testRotErr = zeros(numTests, 1);

for test = 1:numTests

    % theta5 is kept away from 0 and +/-pi, with a random sign so both
       wrist branches are exercised
    sign5 = 2*(rand > 0.5) - 1;
    qTrue = [ (2*rand-1)*pi ; (2*rand-1)*pi ; (2*rand-1)*pi ; ...
              (2*rand-1)*pi ; sign5*(rand*0.9+0.05)*pi ; (2*rand-1)*pi ];
    DH(:,4) = qTrue;

    Tfk = fkChain(DH);
    T06 = Tfk(:,:,end);

    solutions = ikABB(T06, dims);
```

```
48
49      poseOK = size(solutions, 2) == 8;
50      recovered = false;
51      for k = 1:size(solutions, 2)
52          DH(:,4) = solutions(:,k);
53          Tk = fkChain(DH);
54          posErr = norm(Tk(1:3,4,end) - T06(1:3,4));
55          rotErr = norm(Tk(1:3,1:3,end) - T06(1:3,1:3), 'fro');
56          worstPosErr = max(worstPosErr, posErr);
57          worstRotErr = max(worstRotErr, rotErr);
58          testPosErr(test) = max(testPosErr(test), posErr);
59          testRotErr(test) = max(testRotErr(test), rotErr);
60          if posErr > tol || rotErr > tol
61              poseOK = false;
62          end
63
64          angErr = atan2(sin(solutions(:,k) - qTrue), cos(solutions(:,k) -
                qTrue));
65          if norm(angErr) < 1e-6
66              recovered = true;
67          end
68      end
69
70      % Smallest wrapped-angle distance between any two of the solutions
71      testMinDist = inf;
72      for i = 1:size(solutions, 2)-1
73          for j = i+1:size(solutions, 2)
74              diff = solutions(:,i) - solutions(:,j);
75              testMinDist = min(testMinDist, norm(atan2(sin(diff), cos(diff)
                    )));
76          end
77      end
78      minPairDist = min(minPairDist, testMinDist);
79
80      numPoseOK = numPoseOK + poseOK;
81      numRecovered = numRecovered + recovered;
82      numDistinct = numDistinct + (size(solutions, 2) == 8 && testMinDist >
            distTol);
83  end
84
85  % Wrist-singular test case (theta5 = 0): every returned solution must
        still reach the pose
86  qSing = [0.4; 0.3; -0.5; 0.7; 0; -0.2];
87  DH(:,4) = qSing;
88  Tfk = fkChain(DH);
89  T06 = Tfk(:,:,end);
90  solutions = ikABB(T06, dims);
91  singErr = 0;
92  for k = 1:size(solutions, 2)
93      DH(:,4) = solutions(:,k);
94      Tk = fkChain(DH);
95      singErr = max(singErr, norm(Tk(:,:,end) - T06, 'fro'));
96  end
97  singOK = size(solutions, 2) == 8 && singErr <= tol;
```

```
98
99  fprintf('Random test cases:                          %d\n', numTests);
100 fprintf('All 8 solutions reproduce target pose:     %d / %d\n', numPoseOK,
        numTests);
101 fprintf('True joint angles found among solutions:   %d / %d\n',
        numRecovered, numTests);
102 fprintf('8 distinct solutions (min dist > %.0e):   %d / %d\n', distTol,
        numDistinct, numTests);
103 fprintf('Smallest pairwise solution distance:       %.3e rad\n',
        minPairDist);
104 fprintf('Worst position error:                       %.3e\n', worstPosErr);
105 fprintf('Worst rotation error (Frobenius norm):     %.3e\n', worstRotErr);
106 fprintf('Singular case (theta5 = 0) pose error:     %.3e\n', singErr);
107
108 fig = figure('Color', 'w');
109 semilogy(1:numTests, max(testPosErr, eps), 'o', 'MarkerFaceColor', [0.1
        0.3 0.7], 'MarkerSize', 4); hold on;
110 semilogy(1:numTests, max(testRotErr, eps), 's', 'MarkerFaceColor', [0.85
        0.4 0.1], 'MarkerSize', 4);
111 yline(tol, '--r', 'Tolerance');
112 ylim([1e-16 1e-6]);
113 xlabel('Test case'); ylabel('Worst error over 8 solutions');
114 legend('Position error (m)', 'Rotation error (Frobenius)', 'Location', '
        best');
115 title('Problem 1: IK verification over 100 random test cases');
116 grid on;
117 exportgraphics(fig, '../Latex/Figures/Problem1ErrorFigure.png', '
        Resolution', 200);
118
119 fid = fopen('../Latex/Problem1Results.tex', 'w');
120 fprintf(fid, '\\begin{tabular}{l r}\n\\hline\n');
121 fprintf(fid, 'Random test cases & %d \\\\\n', numTests);
122 fprintf(fid, 'All 8 solutions reproduce target pose & %d / %d \\\\\n',
        numPoseOK, numTests);
123 fprintf(fid, 'True joint angles found among solutions & %d / %d \\\\\n',
        numRecovered, numTests);
124 fprintf(fid, '8 distinct solutions (min pairwise distance $>10^{-6}$ rad)
        & %d / %d \\\\\n', numDistinct, numTests);
125 fprintf(fid, 'Smallest pairwise solution distance (rad) & $%.2e$ \\\\\n',
        minPairDist);
126 fprintf(fid, 'Worst position error (m) & $%.2e$ \\\\\n', worstPosErr);
127 fprintf(fid, 'Worst rotation error (Frobenius norm) & $%.2e$ \\\\\n',
        worstRotErr);
128 fprintf(fid, 'Singular case ($\\theta_5=0$), all solutions reach pose & %s
        \\\\\n', string(singOK));
129 fprintf(fid, '\\hline\n\\end{tabular}\n');
130 fclose(fid);
```

ikABB.m

Listing 2: ikABB.m

```
% ikABB: Closed-form inverse kinematics of the ABB arm.
%
% q = ikABB(T06, dims)
% Returns every joint-angle solution that reaches the pose T06.
%
% q = [6x8] one solution per column, [theta1; ...; theta6] wrapped to (-pi, pi].
%     Columns are ordered by (shoulder flip, elbow, wrist). An unreachable pose
%     returns a 6x0 matrix. At a wrist singularity (sin(theta5) = 0) theta4 is set
%     to 0, so the two wrist columns of each arm configuration are identical.
%
% T06 = [4x4] desired homogeneous pose of frame 6 in the base frame
% dims = [1x5] link parameters [d1, a2, a3, d4, d6]
%
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function q = ikABB(T06, dims)
    d1 = dims(1);  a2 = dims(2);  a3 = dims(3);  d4 = dims(4);  d6 = dims(5);
    clampTol = 1e-9;
    sinTol = 1e-9;

    R06 = T06(1:3, 1:3);
    P06 = T06(1:3, 4);


    P05 = P06 - d6 * R06(:,3);

    r = sqrt(P05(1)^2 + P05(2)^2);
    D = sqrt(r^2 + (P05(3)-d1)^2);
    L = sqrt(a3^2 + d4^2);

    q = zeros(6, 0);
    if D < eps
        return;
    end

    cosBeta = (a2^2 + D^2 - L^2) / (2 * a2 * D);
    cosGamma = (a2^2 + L^2 - D^2) / (2 * a2 * L);

    if abs(cosBeta) > 1 + clampTol || abs(cosGamma) > 1 + clampTol
        return;
    end
    cosBeta = max(-1, min(1, cosBeta));
    cosGamma = max(-1, min(1, cosGamma));
```

```
46
47      beta = atan2(sqrt(1 - cosBeta^2), cosBeta);
48      gamma = atan2(sqrt(1 - cosGamma^2), cosGamma);
49      gammaSupp = pi - gamma;
50      delta = atan2(d4, a3);
51
52      q = zeros(6, 8);
53      k = 0;
54      for flip = [0 1]
55          thetaOne = atan2(P05(2), P05(1)) + flip*pi;
56          rSigned = (1 - 2*flip) * r;
57
58          for elbow = [1 -1]
59              thetaTwo = atan2(P05(3)-d1, rSigned) + elbow*beta;
60              thetaThree = -elbow*gammaSupp - delta;
61
62              % OR3 = Rz(th1) Ry(-(th2+th3))
63              phi = thetaTwo + thetaThree;
64              c1 = cos(thetaOne);  s1 = sin(thetaOne);
65              R03 = [c1 -s1 0; s1 c1 0; 0 0 1] * ...
66                    [cos(phi) 0 -sin(phi); 0 1 0; sin(phi) 0 cos(phi)];
67              R36 = R03' * R06;
68
69              for wrist = [1 -1]
70                  sinFive = wrist * sqrt(R36(1,3)^2 + R36(2,3)^2);
71                  thetaFive = atan2(sinFive, R36(3,3));
72
73                  if abs(sinFive) < sinTol
74                      % Wrist singularity: only theta4 +/- theta6 is
                            determined
75                      thetaFour = 0;
76                      if R36(3,3) > 0
77                          thetaSix = atan2(R36(2,1), R36(1,1));
78                      else
79                          thetaSix = atan2(R36(2,1), -R36(1,1));
80                      end
81                  else
82                      thetaFour = atan2(-R36(2,3) / sinFive, -R36(1,3) /
                            sinFive);
83                      thetaSix = atan2(-R36(3,2) / sinFive, R36(3,1) /
                            sinFive);
84                  end
85
86                  k = k + 1;
87                  qk = [thetaOne; thetaTwo; thetaThree; thetaFour; thetaFive
                        ; thetaSix];
88                  q(:,k) = atan2(sin(qk), cos(qk));
89              end
90          end
91      end
92  end
```

Problem 1 (zero-angle figure)

Listing 3: Problem1Figure.m

```m
% Problem1Figure: Sketch the zero-angle configuration of the ABB arm from
    the given DH table.
%
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% 10/03/2026

clear; close all; clc;

addpath(genpath('../../../Functions'));

d1 = 0.5;  a2 = 1.0;  a3 = 0.2;  d4 = 1.0;  d6 = 0.3;

DH = [      0     d1     pi/2       0   ;
            a2    0      0          0   ;
            a3    0     -pi/2       0   ;
            0     d4     pi/2       0   ;
            0     0     -pi/2       0   ;
            0     0      0          0  ];
DH(6,2) = d6;

T = fkChain(DH);

fig = figure('Name', 'Problem 1: Zero-Angle Configuration', 'Color', 'w');
hold on; grid on; axis equal;
xlabel('X'); ylabel('Y'); zlabel('Z');
title('Zero-Angle Configuration of the ABB Arm');
view(3);

origins = squeeze(T(1:3,4,:))';
plot3(origins(:,1), origins(:,2), origins(:,3), 'k-o', 'LineWidth', 1.5, '
    MarkerFaceColor', 'k');

axisLen = 0.25;
for i = 1:size(T,3)
    p = T(1:3,4,i);
    quiver3(p(1), p(2), p(3), T(1,1,i), T(2,1,i), T(3,1,i), axisLen, 'r',
        'LineWidth', 1.5);
    quiver3(p(1), p(2), p(3), T(1,2,i), T(2,2,i), T(3,2,i), axisLen, 'g',
        'LineWidth', 1.5);
    quiver3(p(1), p(2), p(3), T(1,3,i), T(2,3,i), T(3,3,i), axisLen, 'b',
        'LineWidth', 1.5);
    % Frames 4 and 5 share an origin, so they get one combined label
        offset sideways
    if i == 5
        text(p(1)+0.04, p(2)+0.2, p(3)+0.04, '\{4\},\{5\}');
    elseif i ~= 6
        text(p(1)+0.04, p(2)+0.04, p(3)+0.04, sprintf('\\{%d\\}', i-1));
    end
```

```
45 end
46
47 exportgraphics(fig, '../Latex/Figures/Problem1Figure.png', 'Resolution',
       200);
```

dhTransform.m

dhTransform.m calls the custom `rotZ` *and* `rotX` *functions below (both take radians), not MATLAB's built-in* `rotz/rotx` *(which take degrees).*

Listing 4: dhTransform.m

```
% dhTransform: Homogeneous transform from DH parameters.
%
% H = dhTransform(a, d, alpha, theta)
% Homogeneous transform from DH parameters.
%
% H = [4x4] homogeneous transform
%
% a = link length
% d = link offset
% alpha = link twist (rad)
% theta = joint angle (rad)
%
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% 09/28/2026
function H = dhTransform(a, d, alpha, theta)
    R = rotZ(theta) * rotX(alpha);
    p = [a*cos(theta); a*sin(theta); d];

    H = [R, p;
         0, 0, 0, 1];
end
```

rotZ.m

Listing 5: rotZ.m

```m
% rotZ: Rotation matrix about the Z axis.
%
% R = rotZ(theta)
% Rotation matrix about the Z axis.
%
% R = [3x3] rotation matrix
%
% theta = rotation angle (rad)
%
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% MEGN544
% 09/19/2026
function R = rotZ(theta)
    c = cos(theta);
    s = sin(theta);
    R = [c -s 0;
         s  c 0;
         0  0 1];
end
```

rotX.m

Listing 6: rotX.m

```m
% rotX: Rotation matrix about the X axis.
%
% R = rotX(theta)
% Rotation matrix about the X axis.
%
% R = [3x3] rotation matrix
%
% theta = rotation angle (rad)
%
% Arael Anaya
% 10920967
% MEGN544
% 09/19/2026
function R = rotX(theta)
    c = cos(theta);
    s = sin(theta);
    R = [1 0  0;
         0 c -s;
         0 s  c];
end
```

fkChain.m

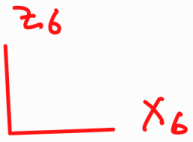
Listing 7: fkChain.m

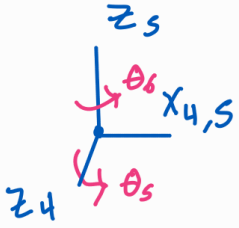
```
% fkChain: Cumulative forward-kinematics transforms from a DH table.
%
% T = fkChain(DH)
% Cumulative forward-kinematics transforms from a DH table.
%
% T = [4x4x(n+1)] stack of transforms, T(:,:,1) = base frame,
%     T(:,:,i+1) = 0Ti for link i
%
% DH = [n x 4] table of [a, d, alpha, theta] rows
%
% Arael Anaya
% 10920967
% MEGN544
% 09/28/2026
function T = fkChain(DH)
    n = size(DH, 1);
    T = zeros(4, 4, n+1);
    T(:,:,1) = eye(4);
    for i = 1:n
        Ti = dhTransform(DH(i,1), DH(i,2), DH(i,3), DH(i,4));
        T(:,:,i+1) = T(:,:,i) * Ti;
    end
end
```

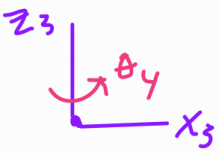
Appendix B: Hand Calculations

a)

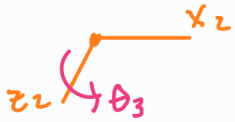
Link	a	d	α	θ
1	0	d_1	$\pi/2$	~~θ~~
2	a_2	0	0	~~θ~~
3	a_3	0	$-\pi/2$	~~θ~~
4	0	d_4	$\pi/2$	~~θ~~
5	0	0	$-\pi/2$	~~θ~~
6	0	d_6	0	~~θ~~

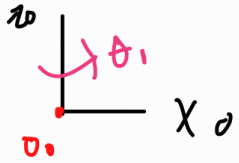






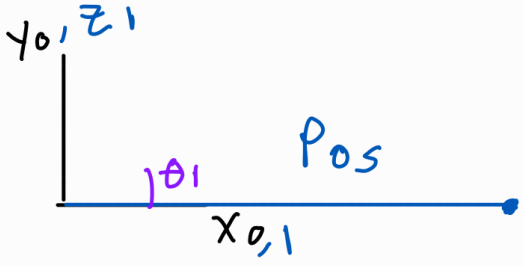






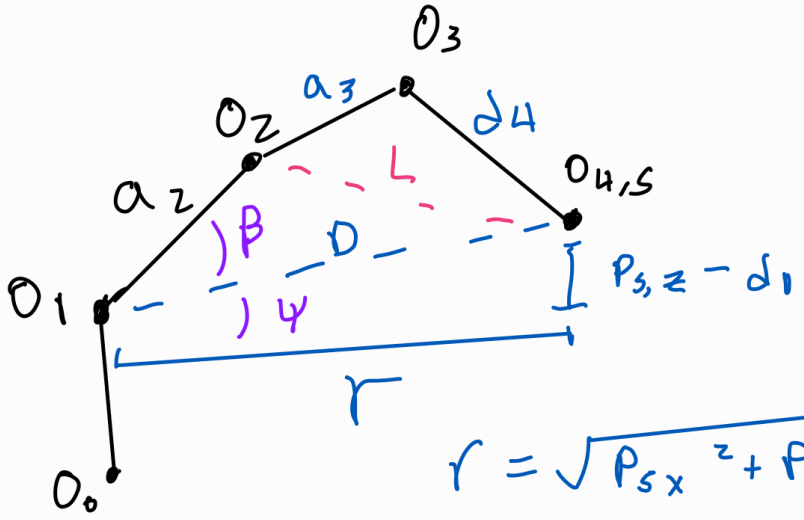
b) Let $P_{0,6} = [P_{6x}, P_{6y}, P_{6z}]^T$

$$P_{0s} = P_{06} - d_6 \cdot R(:,3)$$



or $+\pi$

$$\therefore \theta_1 = \text{atan}(P_{s,y}, P_{s,x})$$



$$r = \sqrt{P_{sx}^2 + P_{sy}^2}$$

$$\psi = \text{atan2}(P_{s,z} - d_1, \sqrt{P_{sx}^2 + P_{sy}^2})$$

$c^2+s^2=1$

$$\theta_2 = \psi \pm \beta$$

$$L^2 = a_2^2 + D^2 - 2a_2 D\, C(\beta)$$

$$D = \sqrt{\sqrt{P_{5x}^2 + P_{5y}^2}^2 + (P_{5z} - d_1)^2}$$

$$L = \sqrt{a_3^2 + d_4^2}$$

$\therefore$

$$C(\beta) = \frac{a_2^2 + D^2 - L^2}{2a_2 D}$$

$$S(\beta) = \pm\sqrt{1 - C^2(\beta)}$$

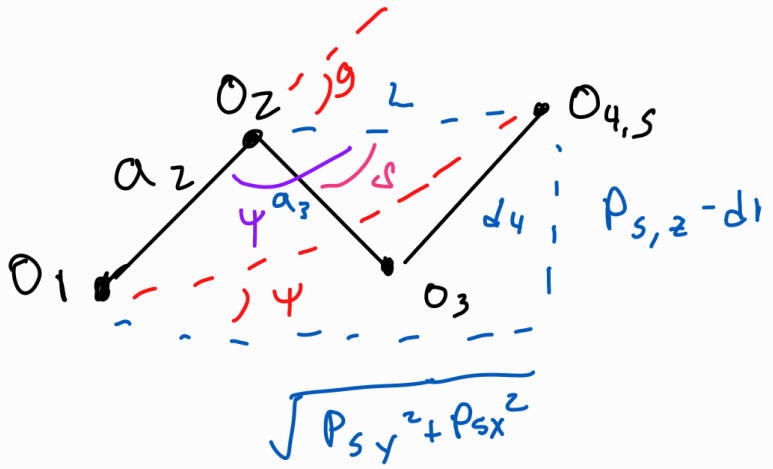
$$\beta = \operatorname{atan2}\left(\pm\sqrt{1 - \frac{L^2 - a_2^2 - D^2}{2a_2 D}},\ \frac{L^2 - a_2^2 - D^2}{2a_2 D}\right)$$

$$\theta_2 =$$

$$\operatorname{atan2}\left(P_{s,z} - d_1, \sqrt{P_{sx}^2 + P_{sy}^2}\right)$$

$$\pm$$

$$\operatorname{atan2}\left(\sqrt{1 - \left(\frac{a_2^2 + D^2 - L^2}{2 a_2 D}\right)^2}, \frac{a_2^2 + D^2 - L^2}{2 a_2 D}\right)$$



$$S = \operatorname{atan2}(d_4, a_3)$$

$$\phi = \pi - \gamma$$

$$\theta_3 = \phi - \delta = (\pi - \gamma) - \delta$$

$$D^2 = a_2^2 + L^2 - 2 a_2 L \, C(\gamma)$$

$$\therefore \; C\gamma = \frac{a_2^2 + L^2 - D^2}{2 a_2 L}$$

$$S\gamma = \sqrt{1 - C\gamma^2}$$

$$\gamma = \operatorname{atan2}\left(\sqrt{1 - C\gamma^2}, C\gamma\right)$$

$$\therefore$$

$$\theta_3 =$$

$$\pi - \operatorname{atan2}\left(\sqrt{1 - \frac{a_2^2 + L^2 - D^2}{2 a_2 L}}, \frac{a_2 + L^2 - D^2}{2 a_2 L}\right)$$

$$- \operatorname{atan2}(d_4, d_3)$$

Find ${}^0R_3$ from DH table

$${}^3R_6 = {}^0R_3^{-1} \cdot {}^0R_6$$

This is ZYZ

$$\therefore$$

$$\theta_5 = \operatorname{atan2}\left(\pm\sqrt{r_{13}^2 + r_{23}^2},\ r_{33}\right)$$

$$\theta_4 = \operatorname{atan2}\left(\frac{-r_{23}}{s\theta_5},\ \frac{-r_{13}}{s\theta_5}\right)$$

$$\theta_6 = \operatorname{atan2}\left(\frac{-r_{32}}{s\theta_5},\ \frac{r_{31}}{s\theta_5}\right)$$